

Basics of FE-Analysis

BK10A6400

Lecture 1: Introduction

Marko Matikainen

LUT University, Finland

June 1, 2026

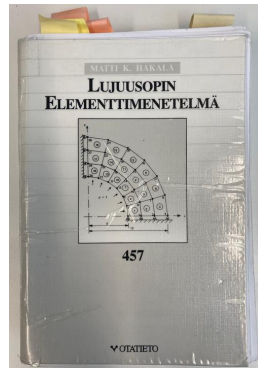
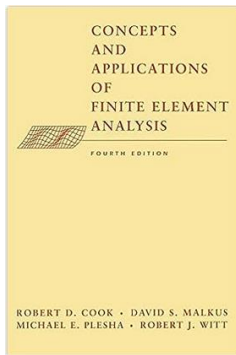
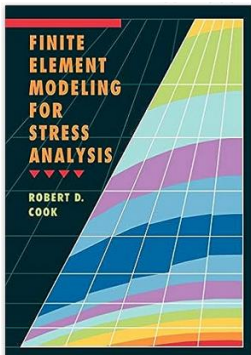
General information

Topic Displacement-based linear finite element method for problems in structural and solid mechanics.

Lectures Marko Matikainen (~~Check times from teaching schedule. Often on Mondays in a class 1316~~)

Exercises Kiia Grönlund (FEMAP) and Uzair Shakeel (ANSYS) (~~every second week, but if needed, every week~~).

- Robert D. Cook: "Finite Element Modeling for Stress Analysis" or "Concepts and Applications of Finite Element Analysis"
- Matti K. Hakala : "Lujuusopin elementtimenetelmä"
- Svein Linge and Hans Petter Langtangen : "Programming for Computations - MATLAB/Octave" [▶ Link](#)
- Claus Führer, Jan Erik Solem and Olivier Verdier: "Computing with Python"
- MATLAB - codes and Python - codes (hopefully)
- Some ANSYS macros
- Some lecture notes (but not hundreds of slides) will be available.
- Learning by doing...also during lectures!
- It is essential to attend the lectures. Guidance for the bonus tasks.
- It is important to participate in the exercises. Guidance on the assignments.



Preliminary requirements

- From the third year onwards.
- Ready to work and learn mathematical-oriented (engineering) stuff. The more time you invest, the more you learn.
- Strength of materials
- Matrix calculus
- Anyone used MATLAB or Python?

Passing the course

- 4 credits (104 hours work)
- Course grade: 0-5, exercises 40% and exam 60%.
- Four exercises a' 10 points
- Exam includes multiple choice questions (60 points).
- Attendance for contact hours is strongly recommended. Be on time!
- During lectures, bonus tasks will be shared and guidance will be given.
- Bonus points (from bonus tasks) will be included in exam points.
- You should earn 50 points to pass the course.
- It is possible to earn more than 100 points due to bonus tasks, but the grade cannot be higher than 5 :-D.

Bonus tasks

Have to return via Moodle before the deadline. No points after it. Usually around a week to do it.

Passing the course

Bonus tasks

Have to return via Moodle before the deadline. No points after it. Usually around a week to do it.

Exam

It is possible to pass the course without an exam by earning 50 points.

Working load

- $4 \text{ ECT} \cdot 26\text{h} = 104 \text{ h}$
- Contact hours (lectures + exercises): $14 \text{ weeks} \cdot 2\text{h} + 13 \text{ weeks} \cdot 2\text{h} = 54\text{h}$
- 50 h for all assignments and exam
- ≈ 3.5 hours per week.

Working load

- $4 \text{ ECT} \cdot 26\text{h} = 104 \text{ h}$
- Contact hours (lectures + exercises): $14 \text{ weeks} \cdot 2\text{h} + 13 \text{ weeks} \cdot 2\text{h} = 54\text{h}$
- 50 h for all assignments and exam
- $\approx 3.5 \text{ hours per week.}$

Average student's working load excluding contact hours

$\approx 3.5 \text{ hours per week}$

- Solving small problems in structural and solid mechanics with FEM using MATLAB
- You must use the code templates shared in the course.
- Everybody has own initial parameters for the problem.

myFirstFE in MATLAB

The screenshot displays the MATLAB R2022a environment. The Editor window on the left contains the script `run_myFirstFE.m` with the following code:

```

1 clear all;
2 close all;
3 clc;
4
5 % Now we and fpa
6 L=2000; % Length of structure (mm)
7 E=210000; % Young's modulus (MPa)
8 D=100; % Diameter (mm)
9 A=pi*D^2/4; % Area (mm^2)
10
11 % Define stiffness matrix
12 K=E*A/L*[1 -1;
13          -1 1];
14
15 % Define applied force (N)
16 F=[0, 10000];
17
18 % Add boundary conditions
19 K(1,2)=0;
20 F(1)=0;
21
22 % Solve displacements
23 ured=K\F;
24
25

```

The Command Window on the right shows the execution results:

```

>>
used =
    0.0121

fj >>

```

The Workspace window on the right shows the following variables:

Name	Value
A	7.8540e+03
D	100
E	210000
F	[0, 10000]
K	10000
Kred	[8.2467e+05, -8.2467e+05]
L	2000
ured	0.0121

The status bar at the bottom indicates the zoom level is 100%, the current file is 'run_myFirstFE.m', and the cursor is at line 25, column 1.

After having passed the course, the student can:

- understand the basic theoretical background of the displacement-based FEM
- solve displacements (also strains and stresses) about statically loaded mechanical structures.
- solve eigenfrequencies and modes of mechanical structures.
- use FE-analysis software for simple mechanical structure analysis

The purpose of the lectures is to give basic knowledge about

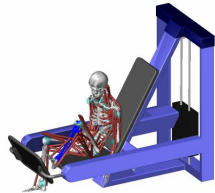
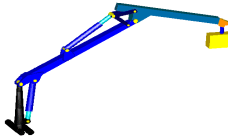
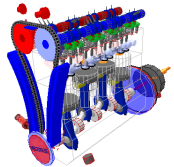
- elemental stiffness matrix
- assembling the global stiffness matrix
- boundary conditions and loads
- solution and source of errors
- linear static and linearized dynamics analysis with FEM

FE analysis using the commercial finite element software

FEMAP will be introduced in the exercises.

Motivation

- Displacements, strains and stresses in structures during static and/or dynamical loading
- Gain a better understanding of loadings that lead to too large deflections or even failure.



Why should you know the FEM deeper than just being a dummy user of FEA packages?

- to know the limitations of finite elements.
- to understand the source of errors.
- to choose a proper solution method (linear, nonlinear, time-dependent,..) for your problem
- to feel great after understanding difficult stuff.

Why should you know the FEM deeper than just being a dummy user of FEA packages?

- to know the limitations of finite elements.
- to understand the source of errors.
- to choose a proper solution method (linear, nonlinear, time-dependent,...) for your problem
- to feel great after understanding difficult stuff.

The most important thing to learn FEM is to know

the right way to set up your problem.

Why should you know the FEM deeper than just being a dummy user of FEA packages?

- to know the limitations of finite elements.
- to understand the source of errors.
- to choose a proper solution method (linear, nonlinear, time-dependent,...) for your problem
- to feel great after understanding difficult stuff.

The most important thing to learn FEM is to know
the right way to set up your problem.

Warning!

Computer programs follow the philosophy of GIGO!

What is FEM?

▶ Link

- Generally: Numerical method to solve a set of partial differential equations.
- In the course: Numerical method to solve displacements of structures.
- Often an important part of multiphysics software.

▶ Link

Examples from SimScale <https://www.simscale.com/docs/tutorials/>

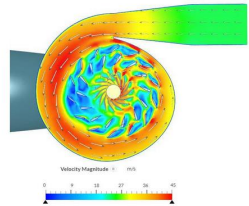
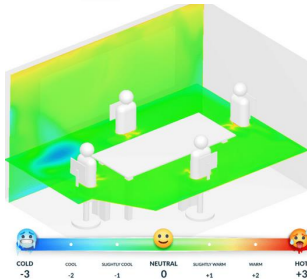
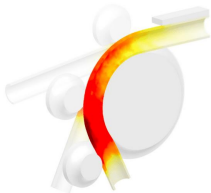
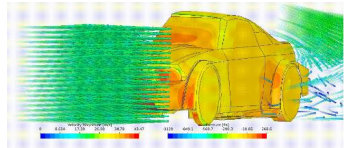
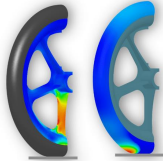
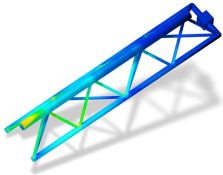
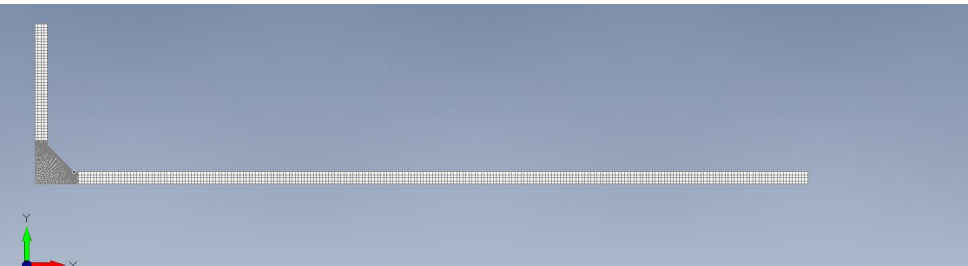
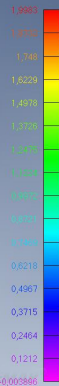


Figure 1: Velocity vectors plotted onto the velocity field.

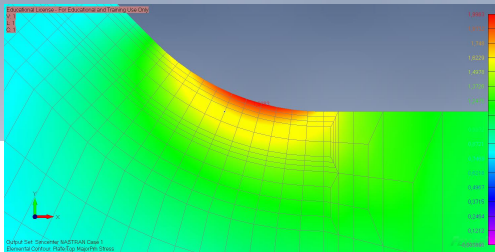


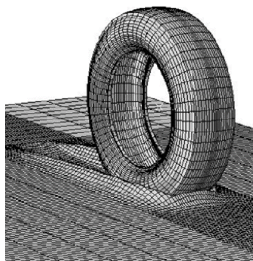
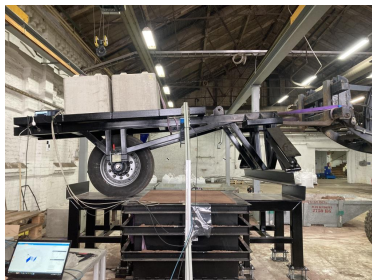
Educational License - For Educational and Training Use Only

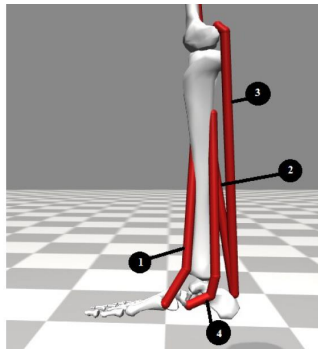
V: 1
L: 1
C: 1



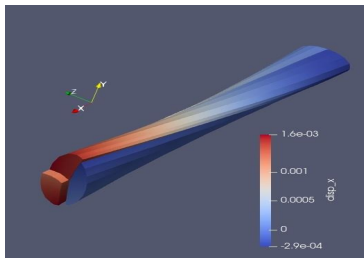
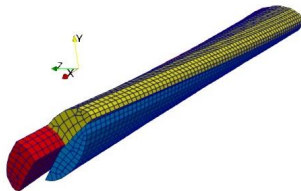
Output Set: Simcenter NASTRAN Case 1
Elemental Contour: Plate Top MajorPtn Stress







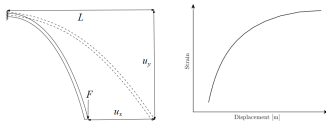
- Muscles:
- | | |
|-----------------------|-----|
| 1. Tibialis anterior | TA |
| 2. Soleus | SOL |
| 3. Gastrocnemius | GA |
| 4. Tibialis posterior | TP |



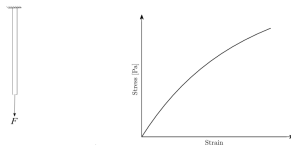
- Numerical (approximative) method for solving partial differential equations (PDE).
- Problems in structural analysis, fluid flow, electromagnetic etc.
- Originally developed by engineers for structural analysis in Aerospace engineering in 1950's.
- The idea is to divide the complicated structure (domain) into a number of simple geometrical elements.
- The primary field (such as displacement) of a single element is interpolated between nodes using simple polynomials.
- Elements are usually connected via common nodes.
- In structural analysis, primary field variables can be (nodal) displacements, stresses
- Displacement-based FEM

$$\mathbf{u} = \mathbf{K}^{-1} \mathbf{f}_{ext}$$

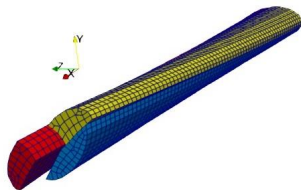
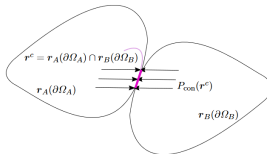
Geometric: Nonlinear strain-displacement



Material: Nonlinear stress-strain

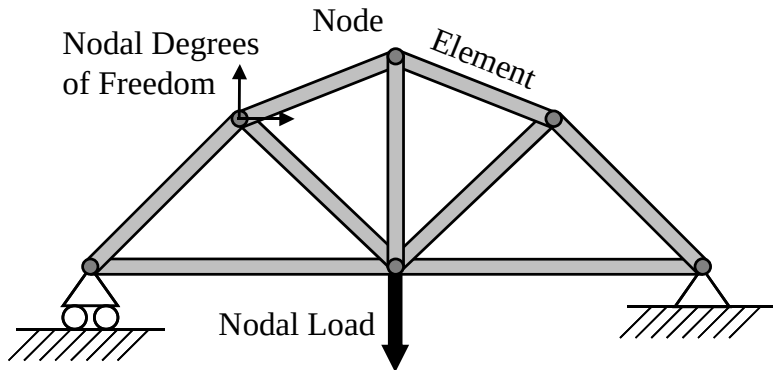


Contact: Nonlinear constraint over boundary



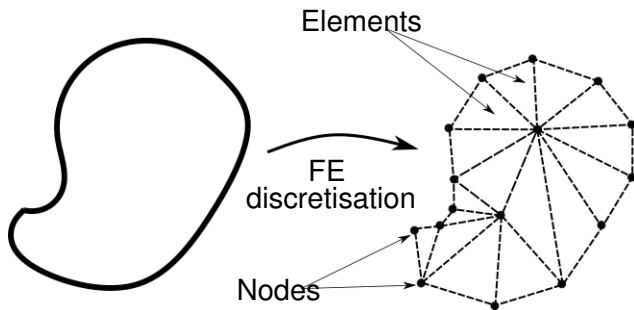
Schematic FE discretisation of an arbitrary 1D structure

- Rod (truss) -type structure
- Beam-type structure



$$\mathbf{u} = \mathbf{K}^{-1} \mathbf{f}_{ext}$$

Schematic FE discretisation of an arbitrary 2D structure



$$\mathbf{u} = \mathbf{K}^{-1} \mathbf{f}_{ext}$$

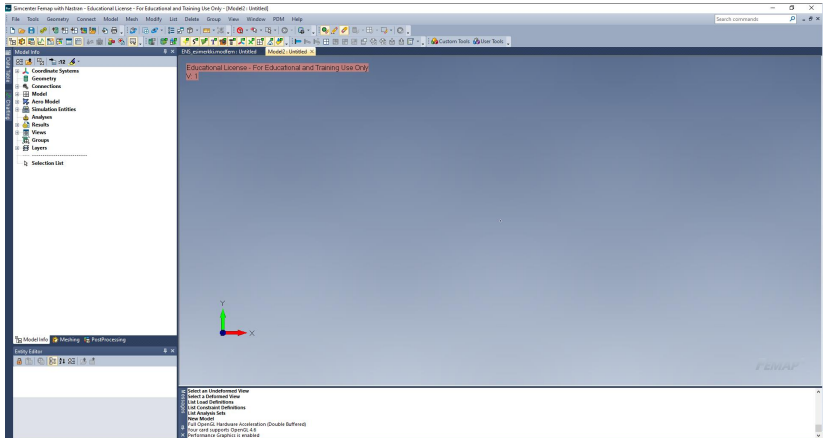
More applications

▶ Link

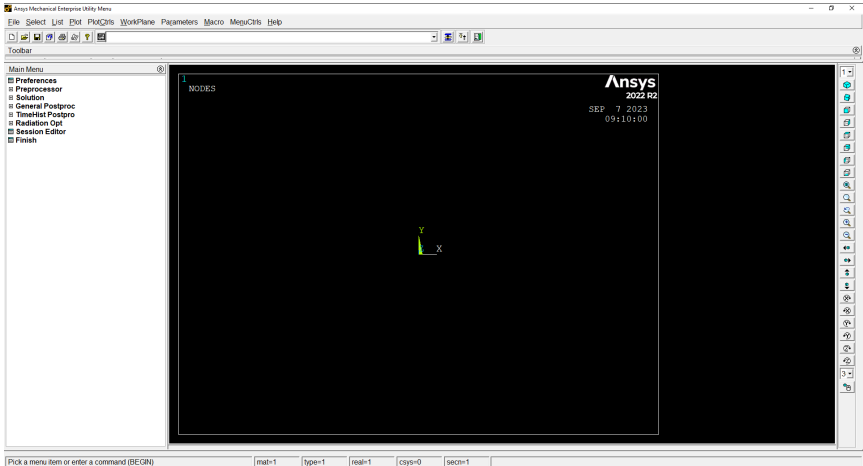
FE softwares

- Preprocessor
 - Element type
 - Create mesh
 - Assembling
 - Material
 - Boundary conditions
 - Loading
- Solver
 - Solution type
- Postprocessor
 - Visualization of displacement, strains and stresses
- Nowadays, many commercial FE softwares (ANSYS, ABAQUS, SimScale, Nastran etc) exist as well as open source FEA packages such as Elmer (developed by Finnish Ministry of Education's CSC) and Salome Meca/Code Aster.
- A commercial FE software FEMAP (pre- and post-processing) is used in exercises. It uses NX/Nastran as solver

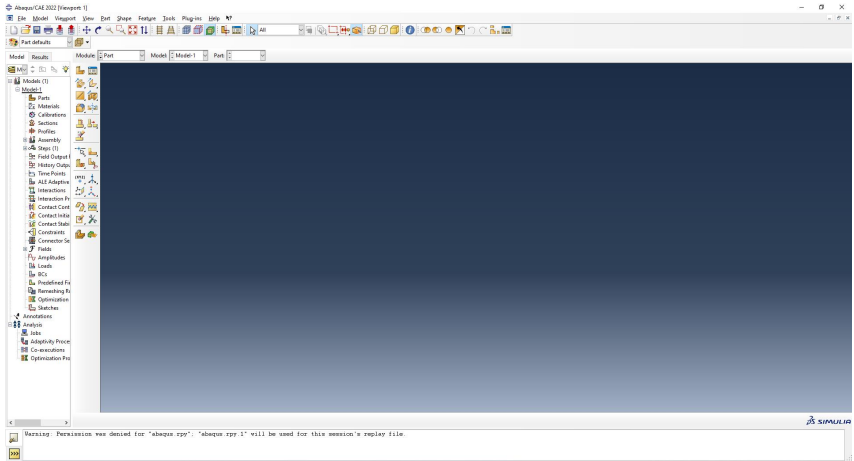
FEMAP graphical user interface



ANSYS graphical user interface



ABAQUS graphical user interface



myFirstFE in MATLAB

The screenshot displays the MATLAB R2022a environment. The Editor window on the left contains the script `run_myFirstFE.m` with the following code:

```

1 clear all;
2 close all;
3 clc;
4
5 % Now we and fpa
6 L=2000; % Length of structure (mm)
7 E=210000; % Young's modulus (MPa)
8 D=100; % Diameter (mm)
9 A=pi*D^2/4; % Area (mm^2)
10
11 % Define stiffness matrix
12 K=E*A/L*[1 -1;
13          -1 1];
14
15 % Define applied force (N)
16 F=[0, 10000];
17
18 % Add boundary conditions
19 K(1,1)=0;
20 K(2,2)=0;
21
22 % Solve displacements
23 ured=K\F;
24
25

```

The Command Window on the right shows the execution results:

```

>>
used =
    0.0121
fj >>

```

The Workspace window on the right shows the following variables:

Name	Value
A	7.8540e+03
D	100
E	210000
F	[0, 10000]
K	10000
Kred	[8.2467e+05, -8.2467e+05]
L	2000
ured	0.0121

The status bar at the bottom indicates the zoom level is 100%, the current file is 'run_myFirstFE.m', and the cursor is at line 25, column 1.

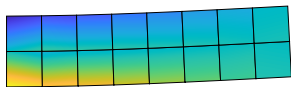
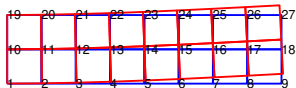
Needs for the numerical solution method

Any numerical solution method for engineering problems should be reliable and efficient.

- Applicable
- Accurate
- Stable
- Inexpensive

Mesh, elements, nodes and degrees of freedom

- Degrees of Freedom, element, nodes, nodal displacements and displacement field.
- Elements are interconnected via nodes.
- There exist hundreds of different element types.
- The most common element types in the structural analysis are rod (truss), beam, shell and solid elements.
- As an example, a simple cantilever problem is analyzed by a four-node plane elements.



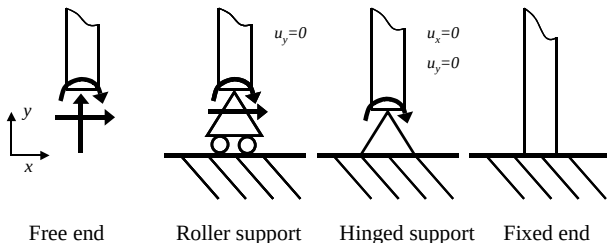
Boundary conditions: constraints and loadings

- If bc:s are not defined correctly, the solution is not a solution for the problem, or the solution cannot be found.
- Loadings in displacement-based FE can be force, pressure, and temperature.
- BC:s and loadings can be applied to points, edges, and surfaces of the objects, but they are always distributed on nodes in the code.

Boundary conditions

The degrees of freedom (DOF) can be divided into translations and rotations. DOF's can be either free or fixed.

- Free end
- Roller support
- Hinged support
- Fixed end



Recap on the basics of Strength of Materials (one-dimensional case)

Strain:

$$\varepsilon = \frac{\delta}{L}$$

Displacement:

$$\delta = L_1 - L$$

Stress (Hooke's law):

$$\sigma = E\varepsilon$$

Stress can also be defined as:

$$\sigma = \frac{F}{A}$$

Substitute strain and stress into Hooke's law:

$$\delta = \frac{FL}{EA}$$

The procedure of linear FEM

- 1 Formulation of elemental stiffness matrices
 - Derive (or use ready solution) local elemental stiffness matrix $\overline{\mathbf{K}}$
- 2 Assembling global stiffness matrix and a vector of external (global) forces for a structure.
 - Compute global elemental stiffness matrix for element k :
$$\mathbf{K}^{(k)} = \mathbf{T}^T \overline{\mathbf{K}}^{(k)} \mathbf{T}.$$
 - Assembling global stiffness matrix $\mathbf{K}_{sys}$
 - External force vector of a structure $\mathbf{f}$
- 3 Boundary conditions (and global force vector): $\mathbf{K}_c, \mathbf{f}_c$
- 4 Solving unknown (nodal displacements) from a system of linear equations $\mathbf{u}_c = \mathbf{K}_c^{-1} \mathbf{f}_c$
 - Inverse of stiffness matrix!
 - Direct solvers
 - Iterative solvers
- 5 Solving forces, strains and stresses. Visualizations.

Stiffness matrix of a rod/truss element

$$\overline{\mathbf{K}} = \frac{EA}{L} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad (1)$$

- Symmetric
- Determinant: $\det \overline{\mathbf{K}} = 0$
- A stiffness matrix of a rod element has one rigid body mode (in axial direction).
- Matrix rank: $\text{rank } \overline{\mathbf{K}} = 1$

MATLAB: Solve displacements of a simple rod under axial loading using one rod (truss) element only.